Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
	(b)			**Mass spectrum** molecular ion peak(s) ⇨ M_r = 92/94 (1) ratio of peak heights 3:1 for molecular ion peaks ⇨ Cl present (1) [accept also for atomic peaks at 35/37] award (1) for any correct fragment on mass spectrum e.g. peak at 15 ⇨ CH_3 peak at 29 ⇨ C_2H_5 **^{13}C NMR spectrum** 4 peaks ⇨ 4 carbon environments (1) molecular formula ⇨ C_4H_9Cl (1) **^{1}H NMR spectrum** 4 peaks ⇨ 4 hydrogen environments (1) award (1) for any one peak identified peak at 4.0 ⇨ **H**C—Cl peak at 1.5 ⇨ —C**H$_3$** peak at 1.3 ⇨ R—C**H$_2$**—R peak at 0.9 ⇨ R—C**H$_3$** award (1) for number of hydrogens in each environment linked to peak heights award maximum of 7 out of these 8 marks final mark reserved for correct structure of **X** ⇨ 2-chlorobutane (1)	1 1	3	3	8		
				Question 11 total	**6**	**3**	**3**	**12**	**0**	**4**